

(7.175)—(7.179)

$$\Delta G_0^{P_0} = \frac{(R + h_P)^2}{R^2} \Delta G_P - \frac{(R + h_P)[(R + h_P)^2 - R^2]}{4\pi} \int_{\omega} \frac{\Delta G_0 - \Delta G_0^{P_0}}{l^3} d\omega$$

$$\frac{(R + h_P)^2}{R^2} = 1 + \frac{2h_P}{R}$$

$$\frac{(R + h_P)[(R + h_P)^2 - R^2]}{4\pi} = \frac{R^2}{2\pi} h_P$$

$$l = l_0 \left[1 + \left(\frac{h_P}{l_0} \right)^2 \right]^{1/2}$$

$$\Delta G_0^{P_0} = \left(1 + \frac{2h_P}{R} \right) \Delta G_P - \frac{R^2}{2\pi} h_P \int_{\omega} \frac{\Delta G_0 - \Delta G_0^{P_0}}{l_0^3 \left[1 + \left(\frac{h_P}{l_0} \right)^2 \right]^{3/2}} d\omega$$

$$h \rightarrow kh$$

$$\frac{1}{\left[1 + k^2 \left(\frac{h_P}{l_0} \right)^2 \right]^{3/2}} = 1 - \frac{3}{2} k^2 \left(\frac{h_P}{l_0} \right)^2 + \dots$$

$$\Delta G_0 = \Delta G_{00} + k\Delta G_{01} + k^2\Delta G_{02} + k^3\Delta G_{03} + \dots = \sum_{n=0}^{\infty} k^n \Delta G_{0n}$$

$$\begin{aligned} & \Delta G_{00}^{P_0} + k\Delta G_{01}^{P_0} + k^2\Delta G_{02}^{P_0} + k^3\Delta G_{03}^{P_0} \dots \\ &= \left(1 + \frac{2kh_P}{R} \right) \Delta G_P \\ & - \frac{R^2}{2\pi} kh_P \int_{\omega} \frac{(\Delta G_{00} - \Delta G_{00}^{P_0}) + k(\Delta G_{01} - \Delta G_{01}^{P_0}) + k^2(\Delta G_{02} - \Delta G_{02}^{P_0}) + \dots}{l_0^3} \left[1 \right. \\ & \left. - \frac{3}{2} k^2 \left(\frac{h_P}{l_0} \right)^2 + \dots \right] d\omega \end{aligned}$$

$$\begin{aligned}
\Delta G_{00}^{P_0} + k\Delta G_{01}^{P_0} + k^2\Delta G_{02}^{P_0} + k^3\Delta G_{03}^{P_0} \dots \\
= \Delta G_P + \frac{2kh_P}{R}\Delta G_P - \frac{R^2}{2\pi}kh_P \int_{\omega} \frac{\Delta G_{00} - \Delta G_{00}^{P_0}}{l_0^3} d\omega \\
- \frac{R^2}{2\pi}k^2h_P \int_{\omega} \frac{\Delta G_{01} - \Delta G_{01}^{P_0}}{l_0^3} d\omega - \frac{R^2}{2\pi}k^3h_P \int_{\omega} \frac{\Delta G_{02} - \Delta G_{02}^{P_0}}{l_0^3} d\omega \\
+ \frac{3R^2}{4\pi}k^3h_P \int_{\omega} \frac{\Delta G_{00} - \Delta G_{00}^{P_0}}{l_0^3} \left(\frac{h_P}{l_0}\right)^2 d\omega
\end{aligned}$$

$$\Delta G_{00}^{P_0} = \Delta G_P$$

$$\Delta G_{01}^{P_0} = \frac{2h_P}{R}\Delta G_P - \frac{R^2}{2\pi}h_P \int_{\omega} \frac{\Delta G_{00} - \Delta G_{00}^{P_0}}{l_0^3} d\omega = -\frac{R^2}{2\pi}h_P \int_{\omega} \frac{\Delta G_{00} - \Delta G_{00}^{P_0}}{l_0^3} d\omega$$

$$\Delta G_{02}^{P_0} = -\frac{R^2}{2\pi}h_P \int_{\omega} \frac{\Delta G_{01} - \Delta G_{01}^{P_0}}{l_0^3} d\omega$$

$$\begin{aligned}
\Delta G_{03}^{P_0} \dots &= -\frac{R^2}{2\pi}h_P \int_{\omega} \frac{\Delta G_{02} - \Delta G_{02}^{P_0}}{l_0^3} d\omega + \frac{3R^2}{4\pi}h_P \int_{\omega} \frac{\Delta G_{00} - \Delta G_{00}^{P_0}}{l_0^3} \left(\frac{h_P}{l_0}\right)^2 d\omega \\
&= -\frac{R^2}{2\pi}h_P \int_{\omega} \frac{\Delta G_{02} - \Delta G_{02}^{P_0}}{l_0^3} d\omega + \frac{3R^2}{4\pi}h_P^3 \int_{\omega} \frac{\Delta G_{00} - \Delta G_{00}^{P_0}}{l_0^5} d\omega
\end{aligned}$$